OFDM Transceiver with Channel Estimation and Equalization

MATLAB-Based Digital Communications Simulation

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Project summary

This project simulates a QPSK-based Orthogonal Frequency Division Multiplexing (OFDM) transceiver in MATLAB. The implementation uses 64-point FFT/IFFT processing, a cyclic prefix, a three-tap static multipath channel, additive white Gaussian noise (AWGN), pilot-based least-squares (LS) channel estimation with linear interpolation, and frequency-domain Zero-Forcing (ZF) and Minimum Mean Square Error (MMSE) equalizers. A perfect-channel-knowledge ZF receiver is included as a reference.

This document describes the implementation represented by the project MATLAB script. It does not claim compliance with a specific 5G NR specification or a completed FPGA/RTL implementation.

1. Introduction

OFDM divides a high-rate data stream across multiple orthogonal subcarriers. In a frequency-selective channel, the subcarriers can experience different complex gains. A cyclic prefix helps convert the linear convolution with a sufficiently short multipath channel into an approximately circular convolution over the useful OFDM symbol, allowing frequency-domain equalization when synchronization and other idealized assumptions hold.

2. Objectives

• Build a QPSK-OFDM transmit and receive chain in MATLAB.

• Use 64-point FFT/IFFT processing and a 16-sample cyclic prefix.

• Model a three-tap static multipath channel and add complex AWGN over an SNR sweep.

• Estimate the channel from known pilot subcarriers using LS estimates and linear interpolation.

• Compare pilot-based ZF and MMSE equalization with a perfect-channel-knowledge ZF reference.

• Compute BER at each simulated SNR and visualize the channel frequency response.

3. System Parameters

Parameter	Value / Description
FFT size	64 subcarriers
Cyclic prefix	16 samples
Number of OFDM symbols	2,000
Modulation	QPSK, unit average symbol power
Pilot arrangement	BPSK pilots at every fourth subcarrier (16 pilots)
Data subcarriers	48 subcarriers per OFDM symbol
Channel	Normalized three-tap static complex channel
SNR sweep	0 dB to 24 dB in 2 dB increments
Channel estimation	LS at pilot locations; linear interpolation between pilots
Equalizers	ZF, MMSE, and perfect-CSI ZF reference

4. Transmitter and Channel Model

The script generates random information bits and maps bit pairs to unit-average-power QPSK symbols. The data symbols are placed on non-pilot subcarriers, while known unit-valued BPSK pilots are placed at indices 1, 5, 9, …, 61 (MATLAB indexing). An inverse FFT creates each time-domain OFDM symbol, scaled by √Nfft, and the final 16 samples are prepended as the cyclic prefix.

The channel impulse response is defined as h = [0.9, 0.4e^(j0.5), 0.2e^(-j0.8)] and normalized. Each cyclic-prefix-extended OFDM symbol is convolved with this channel; the cyclic prefix is then removed to form the useful received block. The channel is static across the simulation.

5. Receiver Processing

5.1 OFDM demodulation

After the cyclic prefix is removed in the simulated channel path, the receiver applies a 64-point FFT with normalization by √Nfft to recover the received subcarrier values.

5.2 Pilot-based LS channel estimation

At each pilot location, the received pilot is divided by the known transmitted pilot to obtain an LS channel estimate. Linear interpolation with extrapolation is then used to estimate the channel response on all 64 subcarriers. Pilot estimates contain noise, so the interpolated estimate can differ from the true response, especially where the channel varies rapidly between pilot locations.

5.3 Equalization

The ZF equalizer divides each received subcarrier by its estimated channel response. It can amplify noise when the channel gain is small. The MMSE equalizer uses a regularized coefficient of the form conj(Hest)/(abs(Hest)^2 + noiseVar), balancing channel inversion and noise. The reference receiver uses the known channel response and ZF equalization; it is a simulation baseline rather than a practical receiver that has access to the true channel.

6. Performance Evaluation

The script sweeps SNR from 0 dB through 24 dB in 2 dB steps. At each point, it adds complex Gaussian noise to the channel output, demodulates the OFDM symbols, estimates the channel, applies each equalizer, demaps QPSK symbols, and calculates BER by comparing detected bits with the transmitted bits. It produces a semilog BER comparison plot and a channel-magnitude plot showing the true channel, interpolated estimate, and pilot estimates.

No numerical BER values are reported in this document because the simulation output was not supplied for independent verification. Insert the actual generated figures and measured BER values after running and validating the script.

7. Reproducibility and Running the Code

Save the MATLAB implementation as **ofdm_zf_mmse.m** (or update the filename to match the repository), open MATLAB in that folder, and run:

```
run('ofdm_zf_mmse.m')
```

The script sets a random seed (rng(42)) for reproducibility. The implementation uses standard MATLAB array operations, FFT/IFFT, convolution, interpolation, and plotting functions.

8. Limitations

• The channel is static; time variation, Doppler, carrier-frequency offset, and synchronization errors are not modeled.

• The simulation processes each OFDM symbol separately through the channel rather than passing one continuous concatenated waveform through the channel.

• Linear interpolation is a simple channel-estimation method and may not be optimal for every channel profile.

• The SNR/noise normalization and MMSE noise-variance convention should be checked carefully when interpreting quantitative BER results.

• The project is a MATLAB simulation only; it does not include a verified RTL receiver or FPGA deployment.

9. Possible Future Work

• Validate noise variance and SNR definitions against the chosen signal-power convention.

• Add Doppler and carrier-frequency-offset models and evaluate compensation techniques.

• Compare interpolation methods or pilot arrangements.

• Save BER plots and a table of measured results from a validated run.

• Consider fixed-point modeling and RTL implementation of selected receiver blocks as a separate, verified extension.

10. Conclusion

The project brings together core digital communications and DSP concepts in a reproducible MATLAB workflow: QPSK mapping, OFDM modulation/demodulation, cyclic-prefix processing, multipath/AWGN channel simulation, pilot-based channel estimation, and ZF/MMSE equalization. The BER sweep and channel-response visualization provide a basis for studying receiver behavior. Quantitative conclusions should be added only after checking the simulation conventions and reviewing the actual generated outputs.

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